

Experiment 36

Aim

To recognize a watch from the given image using OpenCV by detecting the circular watch face and its surrounding object region.

Procedure

1. Import OpenCV and NumPy.
2. Read the given watch image.
3. Convert the image into grayscale.
4. Apply Gaussian Blur to reduce noise.
5. Detect circular regions using the Hough Circle Transform.
6. Identify the circular watch dial.
7. Draw a bounding circle around the detected watch.
8. Display and save the recognized image.

Code:

```
import cv2

import numpy as np

img = cv2.imread("watch.jpg")

if img is None:

    print("Error: Image not found!")

    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

gray = cv2.GaussianBlur(gray, (5, 5), 0)
```

```

circles = cv2.HoughCircles(

    gray,

    cv2.HOUGH_GRADIENT,

    dp=1.2,

    minDist=50,

    param1=100,

    param2=40,

    minRadius=40,

    maxRadius=120

)

if circles is not None:

    circles = np.uint16(np.around(circles))

    x, y, r = circles[0][0]

    cv2.circle(img, (x, y), r, (0, 255, 0), 3)

    cv2.circle(img, (x, y), 3, (0, 0, 255), -1)

    cv2.putText(

        img,

        "WATCH",

        (x - 50, y - r - 10),

        cv2.FONT_HERSHEY_SIMPLEX,

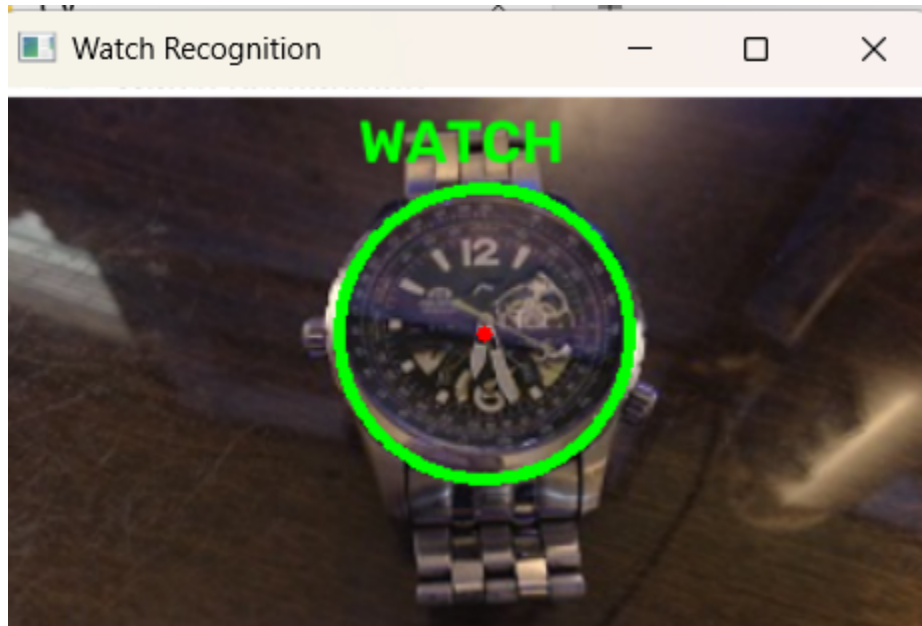
```

```
    0.8,  
    (0, 255, 0),  
    2  
)  
  
print("Object Recognized: WATCH")  
  
else:  
  
    print("Watch not detected")  
  
cv2.imshow("Watch Recognition", img)  
  
cv2.imwrite("watch_recognized.jpg", img)  
  
cv2.waitKey(0)  
  
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The watch was successfully recognized from the given image using OpenCV. The circular watch face was detected using the Hough Circle Transform and marked with a bounding circle and the label "WATCH".